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ARTICLE

Analysis of the Population Structure of *Anopheles funestus* (Diptera: Culicidae) from Western and Coastal Kenya Using Paracentric Chromosomal Inversion Frequencies

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ABSTRACT In total, 324 *Anopheles funestus* Giles specimens collected from seven houses in western Kenya and seven in coastal Kenya were scored for their paracentric chromosomal inversions with the aim of determining the level of genetic differentiation based on these inversions. Houses in each area were within a 2-km radius. The two areas are ≈700 km apart. Only inversions 2a, 3a, 3b, and 5a were found to be polymorphic. Levels of polymorphism varied greatly between inversions. There were no significant deviations from Hardy-Weinberg expectations for samples from individual houses at one site or when data for houses in each area were pooled. Overall, the level of differentiation between western and coastal Kenya was significant, suggesting that the two populations are genetically isolated. Results based on inversion 2a alone were, however, not consistent with this conclusion. Founder effects and selection against the 2a inversion are discussed as possible explanations for this discrepancy.

KEY WORDS *Anopheles funestus*, malaria, chromosomal inversions, variability, Kenya

MALARIA CONTINUES TO be a major health concern in the tropics. *Anopheles funestus* Giles has been shown to be an important malaria vector, in some cases playing a more important role than *An. gambiae* Giles and *An. arabiensis* Patton (e.g., Fontenille et al. 1997). *An. funestus* is a member of a species complex comprising at least nine members, the adults of which are not easily distinguished on the basis of morphological characteristics (Gillies and De Meillon 1968; Gillies and Coetzee, 1987) although some species may be distinguished using larval characteristics. The members of this complex are *An. funestus*, *An. vaneedeni* Gillies & Coetzee, *An. parensis* Gillies, *An. aruni* Sobti, *An. confusus* Evans & Leeson, *An. rivulorum* Leeson, An. *fuscivenosus* Leeson, *An. leesoni* Evans, and *An. brucei* Service. Traditionally, the only other method for distinguishing members of the *An. funestus* group have been by chromosomal inversion karyotypes (Green and Hunt 1980, Green 1982). More recently, however, single-strand conformation polymorphism (SSCP) analysis has been used to identify four member of this group (Koekemoer et al. 1999). Of the nine species in the complex, *An. funestus* has the widest distribution. This mosquito is also highly anthropophilic (Gillies and De Meillon 1968).

Characterization of vector populations in terms of reproductive units is a fundamental step in the development of vector control strategies that are based on the genetic manipulation of vectorial capacity. An understanding of genetic differentiation and gene flow in natural mosquito populations is crucial to determining how far introduced genes will spread, which in turn is important in determining the spatial and temporal scale required for gene introductions. Knowledge of population genetic structure is also useful in predicting the spread of insecticide resistance associated with particular genes.

The characterization of populations in terms of basic reproduction units can be achieved by measuring among-population genetic variability, using any of several genetic markers. In the study reported here, fixed paracentric inversion karyotypes were used to differentiate between the various members of the *An. funestus* group and then polymorphic inversions used to study population genetic structure.

Materials and Methods

Study Site Description. Two sites were used in this study: Ahero in western Kenya (34° 30′ E; 0° 5′ S) and Magaoni in Kwale district of coastal Kenya (39° 30′ E; 4° 10′ S). Each site is traversed by a permanent fresh water stream with vegetation floating along the sides. Several small ponds with tall reeds are also found in both areas. In addition, Ahero is a rice-growing area where irrigation is carried out intermittently. Previous studies carried out in these two areas indicate that *An.*

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funestus plays an important role in the transmission of malaria (Mbogo et al. 1995, Githeko et al. 1996).

Mosquito Collections and Identification. Adult female *An. funestus* group mosquitoes were collected from Magaoni between 3 and 5 May 2000, and from Ahero between 5 and 12 June 2000. Seven houses, all located within a 2-km radius, were sampled per site. Mosquitoes were collected from inside the houses by aspiration and by the pyrethrum spray collection method as in Githeko et al. (1996). Collections were by aspiration from house walls in the morning and by the pyrethrum spray collection method in the afternoons. Adult *An. funestus* complex mosquitoes were distinguished from other anophelines based on morphological characteristics (Gillies and De Meillon 1968, Gillies and Coetzee 1987). Blood-fed mosquitoes collected during the morning were kept under surveillance and preserved individually in Carnoy's fixative in 1.5-ml microcentrifuge tubes once they became half-gravid. Half-gravid females collected by pyrethrum spraying were also preserved in Carnoy's fixative.

Laboratory Procedures. Specimens were removed from Carnoy's fixative, the ovaries dissected, and squash preparations of the polytene chromosomes made according to Hunt (1973). The preparations were examined under phase contrast microscopy (Olympus model BX50, Tokyo, Japan) and paracentric inversion karyotypes were scored using the nomenclature of Green and Hunt (1980) and Green (1982).

Data Management and Analysis. Conformance to Hardy-Weinberg equilibrium was tested in two ways. First, Wright's *F*-statistic was applied (as in Petrarca et al. 2000), where $F = 4ac\text{-}b^2)/[(2a+b)(2c+b)]$ with a and c being the number of the two homozygote classes and b the number of heterozygotes. When the absolute value of F is greater than $(1.96/\sqrt{N})$, there is a significant ($P < 0.05$) departure from expected values. Wright's *F* is a reliable test for sample sizes above 20. The binomial probability test (Sokal and Rohlf 1981) was used to evaluate the significance of multiple tests. The Exact Probability test in GENEPOP version 3.2a (Raymond and Rousset 1995) was also used to test conformance to Hardy-Weinberg expectation and the significance of multiple tests evaluated using Fisher Exact Test. Levels of differentiation were measured using F_{ST} (Weir and Cockerham 1984) in GENEPOP, taking each inversion system as a locus and each alternative chromosomal arrangement to that inversion as an allele. Significance was assessed using the exact test of contingency tables to test for homogeneity of inversion frequencies and of inversion genotypic frequencies across populations. The effective migration rate (*Nm)* was estimated from F_{ST} according to Slatkin's (1987) formulation: $Nm = 1/4[(1/F_{ST})\text{-}1]$. The *Nm* indicates the relative strength of gene flow and genetic drift; observed genetic differentiation is expected to be a result of genetic drift if *Nm* is <1 but if *Nm* is >1, then gene flow overcomes the tendency of genetic drift to cause differentiation (Slatkin 1987). This, however, is not a sharp threshold but a useful characterization of results. Random association of alleles was tested by linkage disequilibrium analysis in GENEPOP.

Results

In total, ≈250 specimens were collected from each of the two areas. Of these, only 158 and 166 specimens were suitable for analysis from Ahero in western Kenya and Magaoni in coastal Kenya, respectively, representing ≈65% of all specimens. Most of the specimens were scored for all inversions. *An. funestus* was the only member of the *funestus* species complex found. No inversions were found on chromosome X and arm 4. Inversions 2a, 3a, 3b, and 5a were polymorphic (Table 1).

Comparison of Mosquito Populations Between Houses. Specimens were first analyzed by house of collection. Only houses with more than twenty specimens were included in the analysis, which included three houses in Ahero and four in Magaoni. Two of the seven houses could not be tested for conformance to Hardy-Weinberg expectations for inversion 2a and two different houses for inversion 3a. For each of these inversions, the sample from one house was fixed for the standard form of the inversion and that from the other house had the expected frequency of the inverted karyotype as zero. There was conformance to Hardy-Weinberg expectations in samples from all other houses for all inversions using both the *F*-statistics method and the Exact Probability Test.

F_{ST} values for the comparison of samples between houses in each of the study sites are given in Table 2. The distribution of inversion frequencies and of inversion genotypic frequencies was not significantly different ($P > 0.05$). N*m* values for all comparisons were >1 (lowest values calculated from mean F_{ST} across all inversions was 8.6; most values were undefined due to the zero F_{ST} values but infinitely large). These results suggest that between-house variability is not significant and that samples from each area represent one interbreeding population.

Comparison Between Mosquito Populations from Ahero (Western Kenya) and Magaoni (Coastal Kenya). For this analysis, data from all the houses (even those with small sample sizes) were pooled because results from the previous analysis suggested that the houses were not differentiated. There was conformance to Hardy-Weinberg expectation for all the inversions in both areas except for inversion 2a for the Ahero population (Exact probability test, $P = 0.0274$; *F*-method, $F = 0.2287$ and $1.96/(N = 0.1645)$. However, when the Fisher exact test for multiple analyses (all four inversions) and the binomial probability test (the probability of getting one of four test being significant when the probability of a significant test is 0.05) were used for the Exact Probability Test method and the *F*-method respectively, this result was found not to be significant (Fisher exact test, $P = 0.2548$ and binomial probability test, $P = 0.1715$).

Inversion frequencies and inversion genotypic frequencies were significantly different between the two

Table 1. Chromosomal inversion frequencies for *An. funestus* from Ahero (in western Kenya) and Magaon (at the Kenyan coast). Wright's *F* and *P* values are also given

Study site	Inversion							
	2 + a		3 + a		3 + b		5 + a	
	Freq.(2n)[a]	*F(P)*	Freq.(2n)	*F(P)*	Freq.(2n)	*F(P)*	Freq.(2n)	*F(P)*
Ahero								
House 1[b]	0.919 (74)	0.2745 (ns)	0.974(78)	−0.0263 (ns)	0.882(76)	−0.1343 (ns)	0.933(90)	0.2857 (ns)
House 2[b]	0.922 (64)	0.3491 (*)	1.000 (68)	−(−)[c]	0.909 (66)	0.1500 (ns)	0.953 (64)	0.3508(*)
House 3[b]	0.957 (70)	−0.04478 (ns)	0.986 (70)	−0.0145 (ns)	0.900 (70)	−0.1111 (ns)	0.923 (78)	−0.08333 (ns)
House 4	0.969 (32)	−0.0323 (ND)	1.000 (32)	−(−)[c]	0.906 (32)	−0.01034 (ND)	0.875 (32)	−0.1429 (ND)
House 5	0.864 (22)	0.6140 (ND)	0.958 (24)	−0.0435 (ND)	0.875 (24)	−0.1429 (ND)	0.708 (24)	−0.0084 (ND)
House 6	0.875 (16)	−0.1429 (ND)	1.000 (24)	−(−)[c]	0.875 (24)	−0.1429 (ND)	0.909 (22)	−0.1000 (ND)
House 7	0.833 (6)	−0.2000 (ND)	1.000 (6)	−(−)[c]	1.000 (6)	−(−)[c]	0.833 (6)	−0.2000 (ND)
All houses	0.926 (284)	0.2287(*)	0.987 (302)	−0.0134 (ns)	0.896 (298)	−0.1161 (ns)	0.908 (316)	0.1064 (ns)
Magaoni								
House 1[b]	0.895 (76)	−0.1176 (ns)	0.690 (84)	0.1088 (ns)	0.571 (84)	0.0278 (ns)	0.313 (80)	−0.2218 (ns)
House 2[b]	0.980 (50)	−0.0204 (ns)	0.776 (58)	−0.0906 (ns)	0.685 (54)	0.0556 (ns)	0.397 (58)	0.2075 (ns)
House 3	0.889 (18)	−0.1250 (ND)	0.889 (18)	−0.1250 (ND)	0.833 (18)	−0.2000 (ND)	0.611 (18)	0.7662 (ND)
House 4[b]	0.964 (56)	−0.1875 (ns)	0.786 (56)	−0.2727 (ns)	0.648 (54)	−0.2180 (ns)	0.375 (56)	0.1619 (ns)
House 5[b]	1.000 (44)	−(−)[c]	0.818 (44)	−0.2222 (ns)	0.750 (44)	0.4370(*)	0.386 (44)	−0.0545 (ns)
House 6	0.813 (16)	−0.2308 (ND)	0.714 (28)	−0.4000 (ND)	0.636 (22)	−0.1786 (ND)	0.375 (16)	−0.0667 (ND)
All houses	0.935 (292)	−0.0700 (ns)	0.759 (332)	−0.0869 (ns)	0.649 (316)	−0.−0.0693 (ns)	0.395 (314)	0.0671 (ns)

ns, *F* value not significant; *, *F* value significant; ND, Significance of *F* not determined because sample size was smaller than 20.
[a] In brackets are the total number of chromosomal arrangements scored for particular inversions.
[b] Houses with more than twenty specimens.
[c] *F* not calculated because sample was fixed for the standard form of the inversion.

areas for inversions 3a, 3b, and 5a ($P \ll 0.001$ for both inversion frequencies and inversion genotypic frequencies). F_{ST} and N*m* values for the comparison between the two areas are given in Table 3. N*m* values for inversions 3a, 3b, and 5a were low (below one for two inversions and just above one for the other). The results based on these inversions therefore suggest that the two populations are differentiated. The distribution of inversion frequencies for inversion 2a were, however, not significantly different between the western and coastal Kenya populations ($P = 0.7430$). The F_{ST} value based on this inversion deviated greatly from the general pattern observed for the other inversions. However, when values were averaged across all inversions, results were significant (homogeneity of inversions and inversion genotype frequencies, $P \ll 0.001$; N$m = 0.66$), suggesting an overall significant level of differentiation between *An. funestus* populations in western and coastal Kenya. No significant linkage disequilibrium was noted, suggesting lack of evidence for natural selection in favor of certain heterozygous genotypes.

Discussion

The habitats described here provide excellent breeding sites for members of the *An. funestus* group as previously described (Gillies and De Meillon 1968). All specimens collected were *An. funestus.* This member of the group is the only one that has highly domestic feeding and resting behavior, with a few other species being found in houses only occasionally (Gillies and De Meillon 1968, Hargreaves et al. 2000). It would therefore be expected that all, or most, specimens collected inside human dwellings would belong to this species; examination of chromosomal species-specific banding arrangements confirmed this expectation.

There was no significant departure from Hardy-Weinberg expectations when chromosomal polymor-

Table 2. F_{ST} values for the comparison between houses in Ahero (in western Kenya) and Magaoni (at Kenyan coast)

Inversions	2a	3a	3b	5a	All inversions
Ahero					
Houses 1 and 2	−0.0196	0.0106	−0.0088	−0.0125	−0.0119
Houses 1 and 3	−0.0041	−0.0103	−0.0106	−0.0127	−0.0096
Houses 2 and 3	−0.0072	−0.0004	−0.0131	−0.0059	−0.0090
Magaoni					
Houses 1 and 2	0.0388	0.0029	0.0112	0.0008	0.0082
Houses 1 and 4	0.0201	0.0080	−0.0023	−0.0060	0.0016
Houses 1 and 5	0.0701	0.0234	0.0496	−0.0030	0.0281
Houses 2 and 4	−0.0145	−0.0147	−0.0145	−0.0206	−0.0167
Houses 2 and 5	−0.0026	−0.0123	−0.0109	−0.0226	−0.0156
Houses 4 and 5	0.0123	−0.0125	0.0066	−0.0223	−0.0088

F_{ST} values were calculated according to Weir and Cockerham (1984).

Table 3. F_{ST} and N*m* values for the comparison between Ahero in Western Kenya and Magaoni at the Kenyan coast

Houses	F_{ST} values	N*m*
Inversion		
2a	−0.0032	Infinitely large
3a	0.2003	0.998
3b	0.1558	1.356
5a	0.4483	0.3077
All inversions	0.2752	0.6584

Data from all houses in each locality were pooled before analysis. F_{ST} values were calculated according to Weir and Cockerham (1984).

phism data were analyzed both at the house and regional levels at each locality. This suggests that no barriers to random mating within populations exist. Populations were polymorphic for inversions 2a, 3a, 3b, and 5a. These inversions have also been found to be polymorphic in *An. funestus* populations from Burkina Faso and Mali (Boccolini et al. 1994, 1998) and in southern Africa (Green and Hunt 1980). The system of inversions in the Kenyan populations, however, does not appear to be as complex as that in West Africa, at least as far as inversions on chromosome arm 2 are concerned. In Senegal, for example, a total of six inversions on this arm, namely a, b, s, t, u, and z, giving rise to six arrangements a, ab, s, t, au, and tz, have been detected (Lochouarn et al. 1998; Dia et al. 2000). Inversions s, t, and u have also been found in populations from Burkina Faso and Mali (Boccolini et al. 1998). Additional research in Burkina Faso indicates that the taxon *An. funestus* may consist of more than one species based on unique inversion characteristics of two sympatric populations (Costantini et al. 1999). Because chromosomal inversions have been shown to be associated with adaptations to environmental heterogeneities (Coluzzi et al. 1979, Toure et al. 1998), the apparently less complex system of inversions in the Kenyan populations may point to lower temporal and spatial environmental heterogeneities.

The level of variability between houses was not significant, suggesting that the populations located within a 2-km radius represent one panmictic population. This result is not surprising given that *An. funestus* has dispersal capabilities of up to 7 km in downwind flights (Gillies and De Meillon 1968).

Overall, the level of variability between the coastal and western Kenya populations (pooled data) was significant, suggesting that the populations are isolated and may experience only limited levels of gene exchange. Such isolation could be maintained by geographical separation (≈700 km for the populations studied here). However, geographical distance may not constitute the only barrier to gene exchange, as has been found for *An. gambiae* (Lehmann et al. 1996, 1999; Kamau et al. 1998, 1999). The influence of the Great Rift Valley, causing discontinuity of this highly anthropophilic species due to lack of human settlement on the temperate escarpments, is most likely an important factor. The limited level of gene exchange, if indeed it occurs, could be expected as a result of transportation, such as by the trains and buses that ply the route between western and coastal Kenya. The Rift Valley may however not present an insurmountable barrier to gene flow: Petrarca et al. (2000) using inversions to study *An. arabiensis* populations from Sudan found that these populations showed characteristics integrating those of populations westward and eastward of the Rift Valley, though with greater similarities to the former. It may be therefore, that the mosquito populations found on either side of the Rift, be they *An. arabiensis* or *An. funestus*, represent the extreme ends of a continuum of characteristics. This question can be addressed only by sampling along a line transect traversing the Rift Valley. Recent studies on inversion polymorphisms in *An. funestus* in West Africa, however, found differentiation between populations located within much shorter distances than in the current study (Lochouarn et al. 1998, Dia et al. 2000).

Considering individual inversions, however, results based on the inversion 2a are inconsistent with the conclusion of isolated populations for western and coastal Kenya. Several possible explanations may be given. Assuming a monophyletic origin of inversions, the similar frequencies in two areas that are so far removed from one another may be as a result of the founder populations in both areas having similar frequencies that have been maintained through time. It may also be that the founder populations had different frequencies of inversions that have converged over time by chance. Another possible explanation is that the 2a inversion in these populations is not monophyletic and arose independently, eventually attaining similar frequencies. Although most available data on the origin of inversions are consistent with the hypothesis of monophyly (e.g., Garcia et al. 1996 in *An. gambiae* complex), a recent study on inversion "2La" of the *An. gambiae* complex suggests that this hypothesis is not without exception; inversions that appear identical at the level of light microscope examination are not necessarily monophyletic (Caccone et al. 1998). In the absence of similar studies in *An. funestus*, one cannot discount this possibility. A different explanation has to do with the nature of inversions as markers of population variability. Most of the uses of genetic markers rely on the fact that they are selectively neutral. This is not necessarily true. Coluzzi et al. (1979) found specific inversions to be associated with adaptations to environmental heterogeneities in *An. gambiae.* Although data on the adaptive significance of inversions in *An. funestus* is still sparse, recent studies suggest that some inversions may be associated with vectorial capacity. Lochouarn et al. (1998), for example, found an association between the inversions 3a and 3b and vectorial capacity in West African populations. Costantini et al. (1999) also found associations between inversion 2a and possibly 3a with infection to *Plasmodium falciparum* in Burkina Faso, although results were somewhat inconsistent between villages. Results of the current study may suggest selection against the 2a inversion in populations from both coastal and western Kenya due to some disadvantage associated with it. This is a plausible expla-

nation given that both populations were almost fixed for the standard form of the 2a inversion. An alternative explanation for the similarity in frequencies would be that 2+a form of the inversion has some adaptive significance and was being maintained in both populations through positive selection pressure due to similarity in environmental conditions such as larval habitats, temperature and humidity. Additional studies would be required to clarify the mechanism of selection that may be in operation. There was, however, no evidence that there is natural selection in favor of certain heterozygous genotypes for all the inversions that were polymorphic.

The overall conclusion, however, is that *An. funestus* populations from western Kenya and those from the coast are genetically isolated. This is in contrast to results from mitochondrial and ribosomal DNA studies that do not support the hypothesis of reproductively isolated chromosomal taxa within *An. funestus* (Mukabayire et al. 1999). Chromosomal inversions, however, are excellent markers of hybridization, and inter-population mating is clearly obvious where such populations are sympatric. Where populations are allopatric, as in the current study, interpretation of the variation is more difficult. As suggested by the above authors, different DNA markers, such as microsatellite loci (Sinkins et al. 2000), should be used to study these populations.

The implications of limited gene exchange mean that the genes of interest, such as those responsible for refractoriness to infection of the mosquito by *Plasmodium* parasites, or those responsible for insecticide resistance, would not freely move between the two populations. The importance of distance as a factor needs to be further investigated by analyzing Kenyan populations located at varying distances between the two sites. The role of the Rift Valley should be investigated by analyzing populations found on either side of this rift system. A comparison of the results obtained in our study and using microsatellite loci (Sinkins et al. 2000) should allow a more balanced understanding of the population genetic structure of this speies.

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